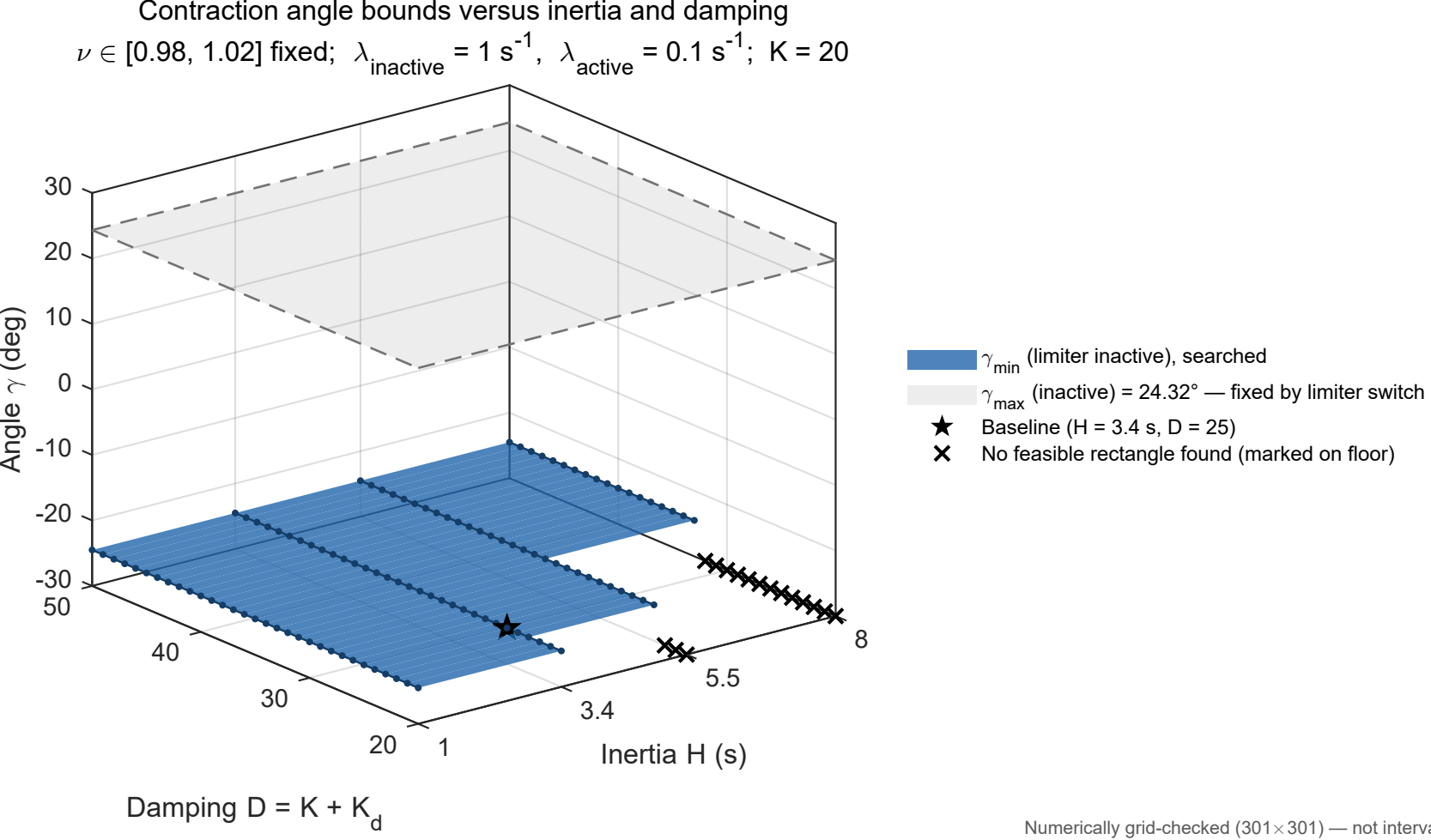

Contraction angle bounds versus inertia and damping
$\nu \in [0.98, 1.02]$ fixed; $\lambda_{inactive} = 1\ s^{-1}$, $\lambda_{active} = 0.1\ s^{-1}$; K = 20
Angle γ (deg)
30
20
10
0
-10
-20
-30
Inertia H (s)
1
3.4
5.5
8
Damping D = K + K_d
50
40
30
20
γ_{min} (limiter inactive), searched
γ_{max} (inactive) = 24.32° — fixed by limiter switch
Baseline (H = 3.4 s, D = 25)
No feasible rectangle found (marked on floor)
Numerically grid-checked (301×301) — not interval-certified